

PIOTTA, Switzerland

WMO: 067530

Lat: 46.5147N

Lon: 8.6881E

Elev: 3333

StdP: 13.01

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.8	17.7	-2.5	5.4	24.5	3.0	7.3	25.5	21.3	34.5	19.3	35.4	4.1	280	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	81.2	63.1	78.3	62.2	75.5	61.1	65.8	76.5	64.2	74.0	62.7	72.0	9.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	94.6	70.3	60.7	89.6	68.4	59.2	85.1	67.1	32.6	76.9	31.2	73.9	30.1	72.2	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.8	14.1	10.5	85.4	3.4	3.8	8.0	88.1	6.0	90.3	4.1	92.4	1.7	95.2		
(5)			9.0	68.6	3.6	2.0	6.3	70.0	4.2	71.2	2.2	72.3	-0.4	73.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	29.9	32.8	40.0	46.4	53.6	61.0	64.3	63.0	55.9	48.3	38.2	31.2		
	DBStd	13.37	6.07	6.38	6.54	6.03	5.62	5.82	4.76	4.79	5.11	5.47	6.04	6.16		
	HDD50	2628	623	482	314	140	27	1	0	0	9	96	354	582		
	HDD65	6632	1088	902	775	559	355	146	70	94	276	517	803	1047		
	CDD50	1583	0	0	4	31	139	332	443	404	185	44	1	0		
	CDD65	111	0	0	0	0	2	26	48	33	2	0	0	0		
	CDH74	926	0	0	0	0	23	259	375	249	20	0	0	0		
	CDH80	140	0	0	0	0	2	39	51	46	2	0	0	0		
(14) Wind	WSAvg	5.4	5.0	5.5	6.1	6.3	6.2	6.2	6.3	5.5	5.1	4.2	4.5	4.5		
(15) Precipitation	PrecAvg	62.8	3.6	3.6	3.7	5.2	6.5	6.1	5.5	6.2	6.1	5.4	6.8	4.0		
	PrecMax	81.7	8.4	9.3	7.8	11.4	17.2	11.6	10.2	9.4	16.8	17.0	22.8	8.5		
	PrecMin	43.3	0.9	0.9	0.5	0.8	2.0	2.4	2.6	1.9	0.9	0.7	1.1	0.4		
	PrecStd	9.8	1.9	2.1	2.0	2.8	3.7	2.4	2.0	1.8	3.9	3.5	4.9	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	54.5	63.6	70.8	77.1	84.2	84.7	85.4	76.5	67.5	57.5	51.5		
		MCWB	39.9	41.2	45.5	52.0	59.0	63.8	62.6	62.9	60.1	53.1	46.7	39.2		
	2%	DB	44.7	48.8	58.9	65.8	72.5	80.2	81.1	79.8	72.4	62.8	52.4	46.0		
		MCWB	37.3	38.4	43.8	49.4	55.7	62.0	63.1	64.0	59.5	52.3	44.1	37.7		
	5%	DB	41.1	45.1	54.8	61.9	68.8	76.9	78.2	76.3	68.6	59.8	49.3	42.6		
		MCWB	34.5	36.4	42.1	47.2	53.8	61.2	62.4	62.6	57.3	51.2	42.8	36.4		
	10%	DB	38.1	41.8	50.8	58.0	65.1	73.5	75.3	73.4	65.5	57.1	46.7	39.7		
		MCWB	33.1	34.7	40.5	45.4	52.1	59.5	61.1	61.2	55.7	50.9	42.2	35.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	42.3	47.4	53.1	60.7	67.1	68.5	67.8	62.9	56.8	49.5	42.6		
		MCDB	47.2	52.1	59.3	67.6	72.8	79.1	79.0	78.6	72.1	61.4	53.4	46.5		
	2%	WB	38.1	39.3	45.1	50.5	57.7	64.5	66.0	65.5	60.9	55.1	46.4	39.7		
		MCDB	42.9	46.9	55.3	62.9	68.6	75.7	76.4	75.8	69.4	59.2	49.9	43.3		
	5%	WB	35.8	37.5	43.2	48.5	55.5	62.5	64.3	64.0	59.1	53.4	44.7	37.6		
		MCDB	39.4	43.2	52.3	58.4	65.4	73.6	73.8	73.1	66.0	57.5	47.3	41.0		
	10%	WB	34.1	35.7	41.4	46.7	53.7	60.7	62.7	62.4	57.4	51.8	43.1	35.8		
		MCDB	36.8	40.8	49.5	55.5	62.0	70.7	71.8	70.5	63.3	55.7	45.9	39.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	12.9	15.4	16.1	16.4	18.2	18.3	17.3	15.6	14.1	10.5	9.5		
		MCDBR	15.0	18.1	22.3	23.9	23.8	24.3	23.2	22.7	20.7	19.0	15.5	14.0		
		MCWBR	9.6	10.7	11.4	11.3	11.5	11.3	10.7	10.9	10.8	10.6	9.6	8.8		
		MCDBR	12.1	15.8	19.2	20.4	19.9	21.5	20.3	19.8	17.9	14.9	12.0	10.5		
	5% WB	MCWBR	8.6	9.9	10.6	10.6	10.8	11.2	10.9	10.5	10.3	9.3	8.5	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.270	0.301	0.354	0.411	0.431	0.442	0.436	0.424	0.393	0.359	0.307	0.267		
	taud		2.272	2.186	2.081	2.025	2.100	2.135	2.160	2.206	2.269	2.323	2.354	2.312		
	Ebn at Noon		261	273	272	266	266	263	262	260	258	249	245	249		
	Edh at Noon		25	33	43	51	49	48	46	42	36	30	23	21		
(44) All-Sky Solar Radiation	RadAvg		510	819	1203	1417	1513	1781	1780	1544	1204	801	497	425		
	RadStd		63	95	174	231	152	159	155	137	118	94	74	55		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.49	+0.94	N/A	N/A	N/A	+104	N/A
Regional (120 neighbors)	N/A	N/A	N/A	+1.31	+0.76	+0.58	N/A	N/A	+103	+27

Nomenclature: See separate page